Organisms and their environment

30 questions: 15 multiple choice, 15 structured.

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1 ? Part 1 Core Energy flow [1]

What is the principal source of energy entering most biological systems?

A soil minerals

B the Sun

C decomposers creating new energy

D carbon dioxide alone

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2 ? Part 1 Core Food chains and food webs [1]

Which organism is a producer in a grassland?

A a rabbit eating grass

B a hawk eating a rabbit

C a fungus decomposing dead grass

D grass making organic nutrients by photosynthesis

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3 ? Part 1 Core Food chains and food webs [1]

In the chain grass → grasshopper → lizard → hawk, what is the lizard?

A secondary consumer

B producer

C primary consumer

D quaternary consumer

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4 ? Part 1 Core Food chains and food webs [1]

Which feeding description identifies a decomposer?

A making glucose directly from sunlight

B eating only living green leaves

C obtaining energy from dead or waste organic material

D absorbing sunlight without using organic material

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5 ? Part 1 Core Food chains and food webs [1]

A mammal eats only other animals. How is it described?

A herbivore

B producer

C photosynthetic organism

D carnivore

6 ? Part 1 Core Food chains and food webs [1]

Why is a food web more informative than one food chain for many habitats?

A It shows that each animal has only one food source.

B It shows interconnected feeding relationships.

C It excludes producers.

D It shows that energy cycles endlessly.

7 ? Part 1 Core Food chains and food webs [1]

Humans remove most predatory fish from a lake. Their main prey are small fish that eat water fleas. What immediate change is most likely, all else unchanged?

A a decrease in small fish caused by more predation

B the extinction of all producers within a day

C an increase in small fish

D conversion of water fleas into producers

8 ? Part 1 Core Food chains and food webs [1]

A single large tree supports thousands of small herbivores. Why can biomass represent this chain better than organism counts?

A It accounts for the large differences in organism size.

B It assumes every organism has the same mass.

C It always measures energy transferred per year.

D It ignores producer material.

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9 ? Part 1 Supplement Food chains and food webs [1]

Which process transfers energy from a consumer to the environment during life?

A photosynthesis in its muscle cells

B respiration releasing energy that is ultimately transferred as heat

C creation of new sunlight

D return of all its energy directly to the producer

10 ? Part 1 Supplement Food chains and food webs [1]

Why do food chains usually contain fewer than five trophic levels?

A Energy increases at every level.

B Producers cannot be eaten.

C Less energy is available at each successive level.

D All organisms are the same size.

11 ? Part 1 Supplement Food chains and food webs [1]

Why can feeding crop plants directly to humans be more energy efficient than feeding them first to cattle?

A Cattle produce extra energy from nothing.

B Humans cannot digest any crop plants.

C Crops have no chemical energy until cattle eat them.

D It avoids energy losses at the cattle trophic level.

12 ? Part 1 Supplement Nutrient cycles [1]

Which process converts nitrogen gas into nitrogen compounds that can enter living systems?

A nitrogen fixation

B denitrification

C evaporation

D combustion of all carbohydrates

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13 ? Part 1 Core Populations [1]

Which description is a population?

A every species in a forest

B all abiotic factors in a pond

C all rabbits living in one meadow at the same time

D all herbivores on Earth at any time

14 ? Part 1 Core Populations [1]

Which group best represents a community?

A all populations of different species in a pond

B all frogs of one species in a pond

C the temperature and oxygen concentration only

D all water molecules in a lake

15 ? Part 1 Core Populations [1]

A new disease spreads in a crowded deer population. What is its likely direct effect on population growth?

A an increase because all deer become producers

B a reduction because more deer die or reproduce less successfully

C no possible effect on survival

D an unlimited increase in food supply

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16 ? Part 1 Core Energy flow [4]

Sunlight enters a grassland food chain.

(a) Describe how the energy first enters living organisms. [2]

(b) Describe its movement through animals and its eventual destination. [2]

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17 ? Part 1

Core

Food chains and food webs

[4]

Pondweed is eaten by snails. Snails are eaten by fish, which are eaten by herons.

(a) State the food chain in the direction of energy transfer. [2]

(b) Identify the producer and tertiary consumer. [2]

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18 ? Part 1

Core

Food chains and food webs

[4]

In a meadow, voles eat grass and seeds; owls eat voles; foxes eat voles and rabbits; rabbits eat grass.

(a) Explain one likely effect on owls if vole numbers fall. [2]

(b) Explain why foxes may be less affected than owls in this food web. [2]

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19 ? Part 1

Core

Food chains and food webs

[4]

A hare eats leaves and a wildcat eats hares.

(a) Explain why the hare is both a consumer and a herbivore. [2]

(b) Explain why the wildcat is a carnivore and state its trophic role in this chain. [2]

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20 ? Part 1

Core

Food chains and food webs

[4]

Fallen leaves are broken down by fungi, while nearby green plants grow.

(a) Describe how the fungi obtain energy and classify their feeding role. [2]

(b) Contrast how green plants obtain organic nutrients and energy. [2]

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21 ? Part 1

Core

Food chains and food webs

[4]

One woodland contains 12 mature trees and 2400 leaf-eating insects. Each tree has far more biomass than one insect.

(a) Explain why numbers alone may give a misleading impression of the producer level. [2]

(b) Explain two advantages of using biomass in this example. [2]

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22 ? Part 1

Supplement

Food chains and food webs

[4]

An ecologist measures annual energy transfer at each trophic level.

(a) Explain why energy transfer is a useful basis for comparing trophic levels. [2]

(b) State two reasons why not all energy in one trophic level reaches the next. [2]

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23 ? Part 1

Supplement

Food chains and food webs

[4]

Plants contain 18 000 kJ of energy available in a measured interval. Herbivore production contains 2700 kJ from that input.

(a) Calculate the percentage transferred into herbivore production. [2]

(b) Explain two reasons why the rest is not stored in herbivore production. [2]

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24 ? Part 1

Supplement

Food chains and food webs

[4]

A farm can produce edible beans or feed the beans to animals and use the animals as food.

(a) Explain why the bean-to-human route can support more people from the same crop energy. [2]

(b) Explain why adding more trophic levels reduces the population supportable at the top. [2]

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25 ? Part 1

Core

Food chains and food webs

[4]

An introduced insect eats the main seeds used by native seed-eating birds. A predator eats those birds.

(a) Explain a possible effect on the native birds. [2]

(b) Explain how the effect could spread to another trophic level. [2]

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26 ? Part 1

Core

Nutrient cycles

[4]

Carbon moves between a forest and the atmosphere.

(a) Describe how carbon enters plants and then herbivores. [2]

(b) Describe two processes that return carbon dioxide to the atmosphere. [2]

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27 ? Part 1

Supplement

Nutrient cycles

[4]

Dead organisms supply nitrogen compounds to soil. Some nitrogen is later lost from waterlogged soil.

(a) Describe how decomposers and nitrifying bacteria make nitrogen available to plants. [2]

(b) Describe nitrogen fixation and denitrification, distinguishing their directions. [2]

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28 ? Part 1

Core

Populations

[4]

A wetland contains reeds, fish, insects and microorganisms interacting with water and soil.

(a) Distinguish the reed population from the wetland community. [2]

(b) Explain why the wetland is an ecosystem. [2]

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29 ? Part 1

Core

Populations

[4]

A rabbit population grows while food is plentiful. Later, food becomes scarce and fox numbers rise.

(a) Explain how food scarcity can limit population growth. [2]

(b) Explain how increased predation can also limit growth. [2]

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30 ? Part 1 Supplement Populations [4]

A bacterial culture initially grows slowly, then rapidly, and finally remains approximately constant in size.

(a) Explain the slow initial phase and the later rapid phase. [2]

(b) Explain why the population can stop increasing. [2]

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